Moule Lin

Marie Skłodowska-Curie Fellow · Ph.D. Candidate, Trinity College Dublin

moulel@tcd.ie · moulelin.github.io · Google Scholar · London, UK (visiting) / Dublin, Ireland

Research Interests

Stochastic processes and probabilistic modelling for large language models · Bayesian deep learning and uncertainty estimation for LLMs · parameter-efficient fine-tuning · knowledge distillation for LLM reasoning and code

Education & Research Positions

Imperial College London 2026.10 – Present
Visiting Researcher, Department of Computing · host: Dr. Yingzhen Li London, UK
– Research on stochastic processes for large language models

Trinity College Dublin 2024.04 – 2028.01
Ph.D. Candidate in Computer Science · supervisor: Dr. Goetz Botterweck Dublin, Ireland
– **Marie Skłodowska-Curie Fellow** (MSCA) — uncertainty estimation for large language models

Eötvös Loránd University 2021.09 – 2023.06
M.Sc. in Artificial Intelligence, Faculty of Informatics Budapest, Hungary
– GPA 3.4/4 · **Stipendium Hungaricum Scholarship** (full scholarship) · **Excellent Thesis** · SH Network Mentor

Northeast Forestry University 2020.09 – 2024.06
M.Sc. in Computer Science, College of Computer and Control Engineering Harbin, China
– GPA 4.38/5 · **National Graduate Scholarship** (2022) · First Class Scholarship for Postgraduates (×3)

Tianjin Agricultural University 2015.09 – 2019.06
B.Eng. in Software Engineering, School of Information and Computer Engineering Tianjin, China
– GPA 3.36/4 · Blue Bridge Cup Algorithm Competition: **National First Prize** (ranked 27 / 51,282) · Tianjin First Prize
– **People's Government Scholarship** (Tianjin) · First Class Academic Scholarship · **Honorary Graduate**

Publications

ICML 2026 — **Lin, M.**, Guan, S., Patane, A., Gregg, D., & Botterweck, G. Bayesian-LoRA: Probabilistic Low-Rank Adaptation of Large Language Models.

AAAI 2026 — **Lin, M.**, Patane, A., Jing, W., Guan, S., & Botterweck, G. Flow-Induced Diagonal Gaussian Processes.

AISTATS 2025 — **Lin, M.**, Guan, S., Jing, W., Botterweck, G., & Patane, A. Stochastic Weight Sharing for Bayesian Neural Networks.

TGRS — **Lin, M.**, Jing, W., Zou, W., Qiu, Z., & Li, C. Gaussian-Based Swap Operator for Context-Aware Extraction of Building Boundary Vectors. *IEEE Transactions on Geoscience and Remote Sensing*.

NeurIPS 2026 — Lan, Q., Fan, M., Zhang, H., Duan, J., **Lin, M.**, Chen, J., Geng, B., & Jiang, X. BrainTRACE: Tracing Longitudinal, Multimodal, and Volumetric Evidence in Brain MRI Clinical Reasoning.

ICLR 2026 — Guan, S., **Lin, M.**, Xu, C., Zhao, J., & Greene, D. Teaching VLMs to Admit Uncertainty in OCR from Lossy Visual Inputs.

EMNLP 2025 — Guan, S., Xu, C., **Lin, M.**, & Greene, D. Effective Synthetic Data and Test-Time Adaptation for OCR Correction. *Main*.

ACL 2025 — Guan, S., **Lin, M.**, Xu, C., Liu, X., Zhao, J., Fan, J., Xu, Q., & Greene, D. PreP-OCR: A Complete Pipeline for Document Image Restoration and Enhanced OCR Accuracy. *Main*.

Industry & Research Experience

Ubotica Technologies — MSCA Secondment
Researcher Ireland
– Deploying uncertainty-aware models on satellites for reliable onboard prediction and decision-making, applying my Bayesian learning and uncertainty estimation research to Ubotica's models

Geely Group — Zeekr 2023.07 – 2023.11
Researcher in Autonomous Driving Shanghai, China
– Researched SOTA BEV models for automatic parking with fisheye cameras and LiDAR; improved large-scale training efficiency by ~8% via prefetched data loading on 2M+ images/videos
– Built a low-light and dirty-camera detection model from data collection to deployment (ONNX, TensorRT) on the Zeekr vehicle platform

Ericsson 2021.07 – 2021.10
Researcher, Model Fusion Budapest, Hungary
– Implemented new algorithms in CUDA/C++ (e.g. a Shift operator to accelerate inference); adapted Transformers into Swin-style local-attention variants for multi-modal data
– Analysed communication-network data to predict user behaviour and preferences, helping operators optimise services

AI Lab, Eötvös Loránd University — Model Compression Project 2022.07 – 2022.10
Researcher Budapest, Hungary
– Researched pruning, distillation and sparse architectures; proposed a novel “Swap” structure with ~10% inference speed-up, co-authoring *Optimized Vectorizing of Building Structures with Swap*